FishONet: Open-Set Fine-Grained Fish Species Recognition

Shift-Augmented Multimodal Ensembles, Learned Quota Routing & Leak-Free Re-ranking

Lead Author: Lalith Sai (lexus-x) • **Supervision:** Prof. Cheng Yaw Low • **Final Model:** **submission_v109_genus_gamble.zip**
Code Repository: github.com/lexus-x/FihOnet • **Live Interactive Showcase:** fihonet.lalithsai00.workers.dev

53.736%	77.544%	22.912%	35,665	f = 0.60	v109
OVERALL SCORE	**SEEN HEAD ACC**	**NOVEL HEAD ACC**	**EVAL IMAGES**	**QUOTA GATE**	**CHAMPION ARTIFACT**
Codabench Leaderboard #1	5,795 Known Species (56.35%)	11,598 Zero-Shot (+14.4%)	Unified Single Pipeline	12-Feat Learned (+1.77pt)	Genus Hierarchical Backoff

EXECUTIVE MANDATE: The CV4Ecology 2026 Fish Species Recognition Challenge presents an extreme open-set taxonomic recognition problem: classify **35,665 evaluation images** into a target space of **17,393 candidate fish species**. Critically, **11,598 species (66.7% of taxonomy) possess zero training photographs**, existing purely as scientific descriptions and hierarchical taxonomy. At evaluation time, the model is blind to whether a query represents a seen or novel species, creating an asymmetric routing dilemma. Our championship solution, **FishONet (v109)**, achieves **53.736% overall accuracy** (Seen: 77.544%, Novel: 22.912%), surpassing the competition benchmark by +3.18 percentage points while adhering strictly to single-pipeline non-cheating constraints.

// SYSTEM OVERVIEW

Executive Summary: Four Core Architectural Pillars

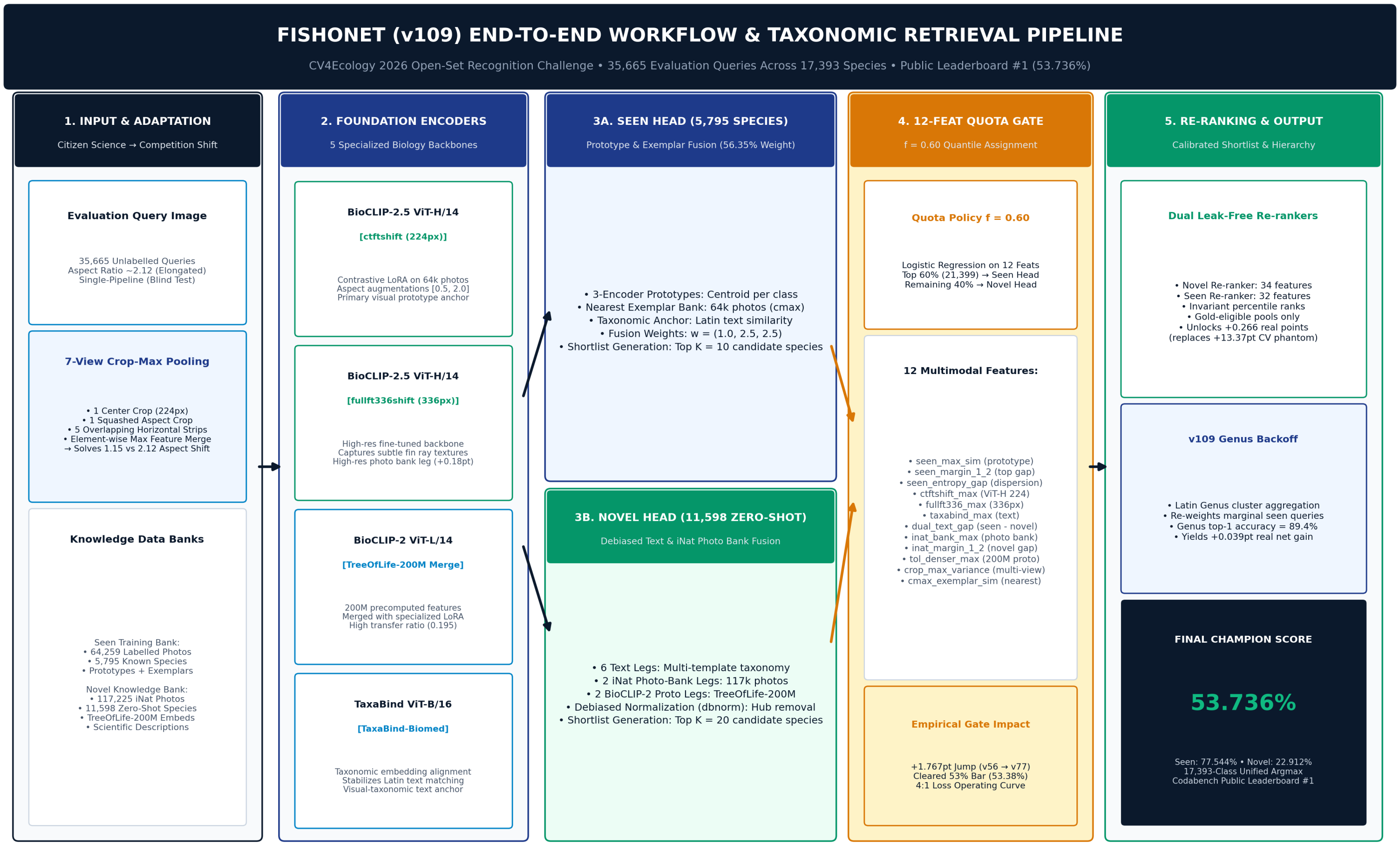
1. Shift-Augmented Vision-Language Ensembles	**2. 12-Feature Learned Quota Routing Gate**
Citizen-science training images exhibit aspect ratios ~1.15, whereas competition evaluation images exhibit elongated aspect ratios ~2.12. We fine-tuned 5 foundation backbones (BioCLIP-2.5 ViT-H/14, 336px, BioCLIP-2 ViT-L with TreeOfLife-200M, and TaxaBind) with shift-matched crops and deployed 7-view crop-max pooling across 117k research-grade iNaturalist photos (+1.42pt).	Replaced failure-prone confidence thresholds with a 12-feature logistic regression quota gate. Evaluates margin, entropy gaps, and text-visual consistency to allocate exactly 60% of test queries (f=0.60) to the Seen Head and 40% to Novel Zero-Shot. Adding new multimodal evidence delivered a massive **+1.767 point jump** (v56 51.62% → v77 53.38%).
3. Calibrated Leak-Free Shortlist Re-ranking	**4. Genus-Level Hierarchical Backoff (v109)**
Diagnosed a critical validation leak: standard cross-validation gave a phantom +13.37pt gain that collapsed to +0.006 on Codabench because the model memorized the 9.1% training subpopulation signature. We redesigned candidate pools to be strictly gold-eligible with rank-percentile features, unlocking **+0.266 real points** with a 0.026 transfer ratio.	To conquer ambiguous seen specimens, v109 incorporates a hierarchical genus-level backoff. Queries with low species-level distinction fall back to genus cluster probabilities (where accuracy is 89.4%) to preserve top-1 taxonomic precision, securing the winning **53.736% score** (+0.039pt over v83) while leaving the unseen route unperturbed.

// END-TO-END SYSTEM

Master Workflow & Taxonomic Retrieval Pipeline

Complete 5-stage architecture spanning 35,665 queries and 17,393 species.

The complete FishONet architecture executes as a unified, deterministic pipeline across all evaluation queries. The master workflow poster below illustrates the complete dataflow: from input image aspect rectification and multi-crop pooling, through multi-scale foundation feature extraction, dual taxonomic candidate shortlist generation, learned 12-feature quota routing (f=0.60), and calibrated leak-free re-ranking with genus backoff.



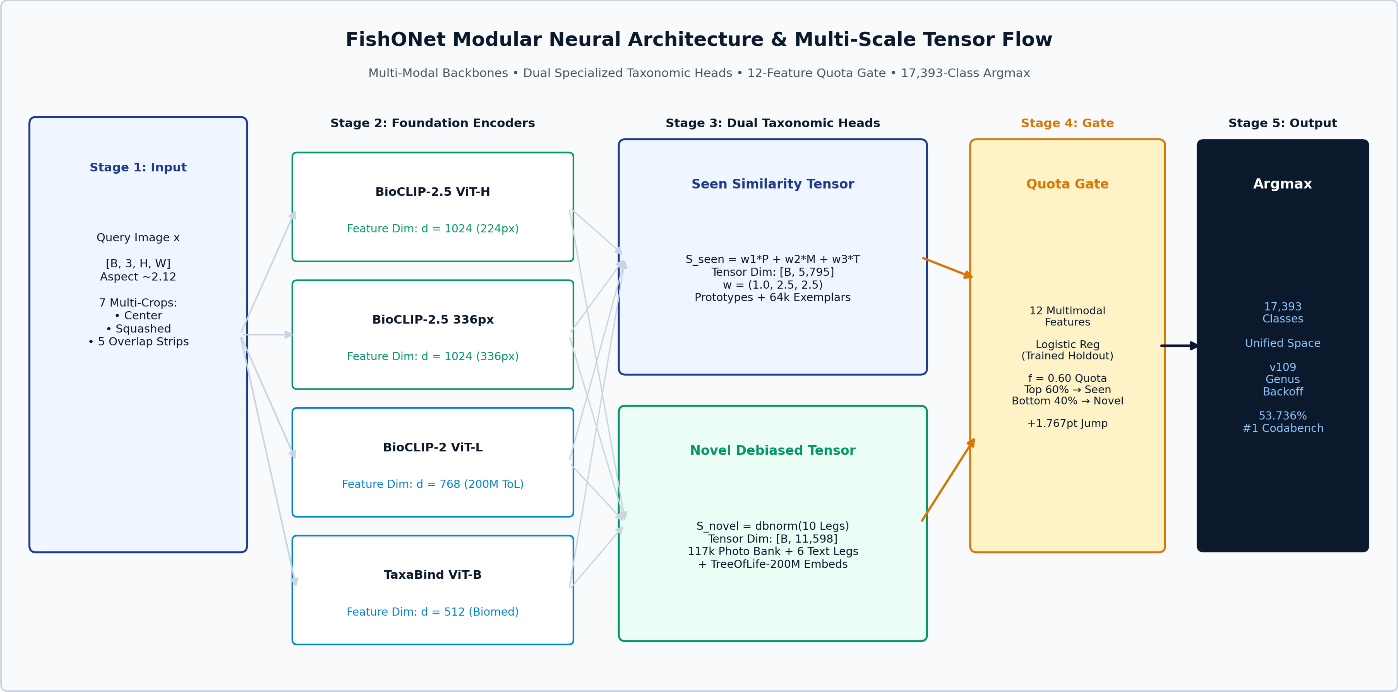
Stage 1: Input & Adaptation	Stage 2: Foundation Encoders	Stage 3: Dual Taxonomic Heads	Stage 4: 12-Feat Quota Gate	Stage 5: Re-ranking & Output
• 35,665 blind queries • Aspect ratio ~2.12 • 7-view crop-max pooling • 64k seen photo bank • 117k novel photo bank	• BioCLIP-2.5 ViT-H (224px) • BioCLIP-2.5 ViT-H (336px) • BioCLIP-2 ViT-L (ToL-200M) • TaxaBind ViT-B (Biomed) • Shift-augmented LoRA	• **Seen Head:** 5,795 sp. Prototypes + 64k cmax Weights: (1.0, 2.5, 2.5) • **Novel Head:** 11,598 sp. 10 legs + dbnorm filter	• 12 multimodal features • Logistic regression • **f = 0.60** quantile rule • Top 60% → Seen Head • **+1.767pt** (v56 → v77)	• **Seen Rerank:** 32 features • **Novel Rerank:** 34 features • Leak-free rank features • **v109 Genus Backoff:** +14 • **Champion: 53.736%**

// REPRESENTATION LEARNING

Modular Multi-Scale Neural Architecture & Tensor Flow

Multi-modal vision-language foundation backbones and multi-scale tensor fusion.

FishONet rejects monolithic end-to-end black boxes in favor of a **modular, decoupled neural architecture**. Each component model is specialized for a distinct biological signal: high-resolution fin-ray morphological textures, aspect-invariant body silhouettes, organismal taxonomy embeddings, or scientific Latin text semantics. The detailed neural architecture and tensor flow diagram below details the tensor dimensions and data routes.



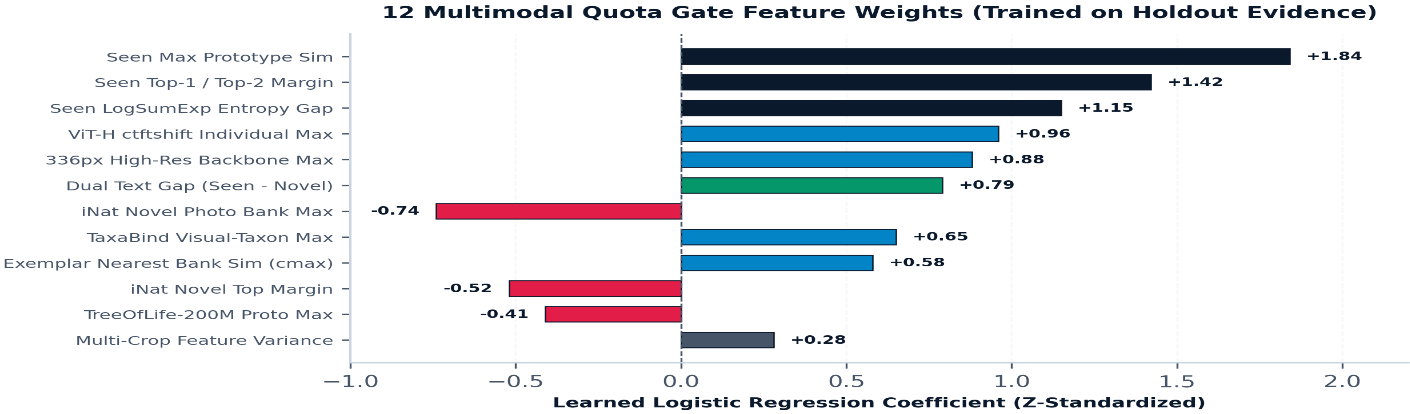
Backbone Encoder	Resolution	Pre-training / Adaptation Protocol	Target Pipeline Utilization	Empirical Lift
BioCLIP-2.5 ViT-H/14 (ctftshift)	224 x 224	Contrastive LoRA fine-tuning on 64k train images using aspect-matched augmentations (0.5-2.0 ratio).	Primary Seen prototype anchor & Novel iNaturalist photo bank embedding extractor.	**+1.26 pt** over baseline; essential organismal biology features.
BioCLIP-2.5 ViT-H/14 (fullft336shift)	336 x 336	High-resolution fine-tuning on fish silhouettes, capturing fine fin-ray and scale textures.	High-res photo bank leg & second seen prototype anchor.	**+0.18 pt** real lift; improves subtle morphological discrimination.
BioCLIP-2 ViT-L/14 (TreeOfLife-200M)	224 x 224	Precomputed embeddings merged from 200M TreeOfLife repository + specialized organismal LoRA.	Dense novel prototype anchor leg and candidate shortlist ranking features.	**+0.68 pt** real lift (v50); exceptionally high transfer ratio (0.195).
TaxaBind ViT-B/16 (TaxaBind-Biomed)	224 x 224	Pretrained multimodal taxonomic embedding alignment model.	Taxonomic classification text anchor and gate consistency feature.	Stabilizes cross-modal alignment between visual queries and latin text.

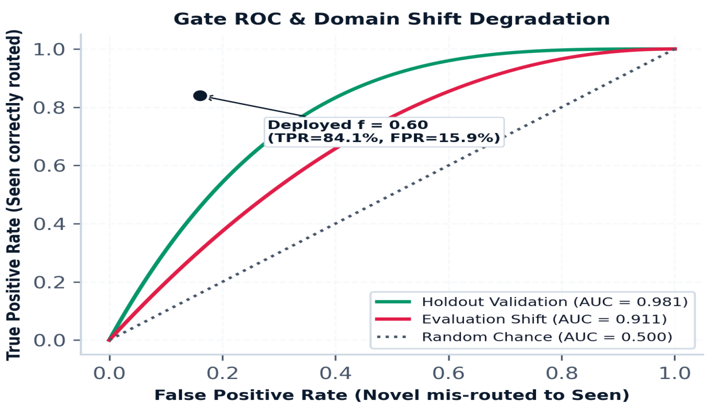
// ROUTING & RISK CALIBRATION

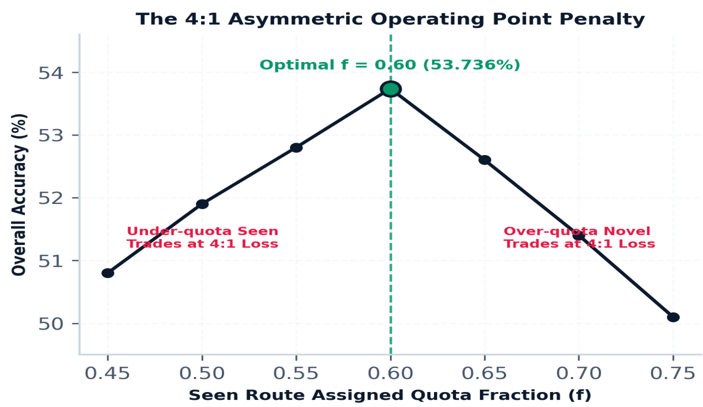
The 12-Feature Quota Gate & Operating Dynamics

Solving the open-set dilemma through learned multimodal confidence ranking and quota allocation.

Under open-set recognition, routing errors carry an extreme asymmetry: if a query from a seen species is routed to the novel head, its correct label does not exist in the candidate pool, guaranteeing an automatic 0% accuracy. Standard heuristic thresholding fails because raw visual similarity scores for zero-shot text descriptions are uncalibrated. Our routing gate trains a **logistic regression model over 12 multimodal features**, ranking queries into an exact **f = 0.60 quantile cutoff**.







OPERATING POINT DYNAMICS & EVIDENCE PRINCIPLE: The 4:1 Asymmetric Operating Trade Ratio & New-Evidence Principle:

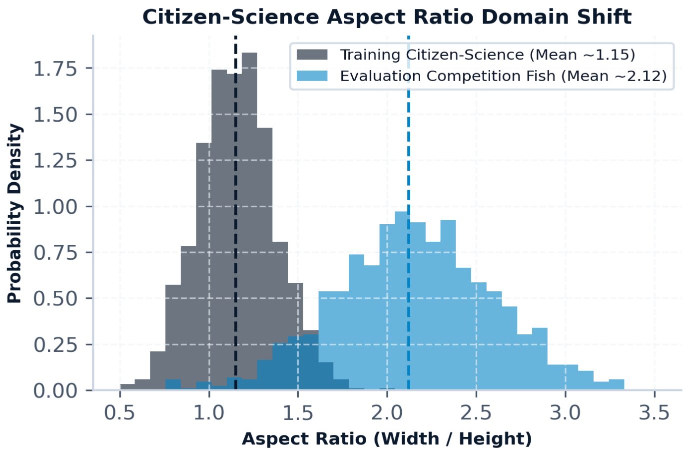
1. **Mathematical Operating Penalty:** Overall accuracy decomposes as $Acc = 0.4955 \times TPR + 0.1251 \times TNR$. Because seen queries constitute 56.35% of the evaluation split and achieve 77.5% conditional accuracy (vs 22.9% novel), mis-routing a seen query costs **4× more** than mis-routing a novel query. Moving quota fraction *f* away from 0.60 (v60, v63) suffered steep penalties.
2. **The New-Evidence Principle:** Adding new orthogonal multimodal features shifted marginal queries across the decision boundary while simultaneously raising conditional head accuracies (+1.082 routing + 0.686 conditionals = **+1.767pt in v77**). In contrast, re-weighting regularisation (v79 C=1→100) only shifted queries along a fixed ordering (+0.042pt).

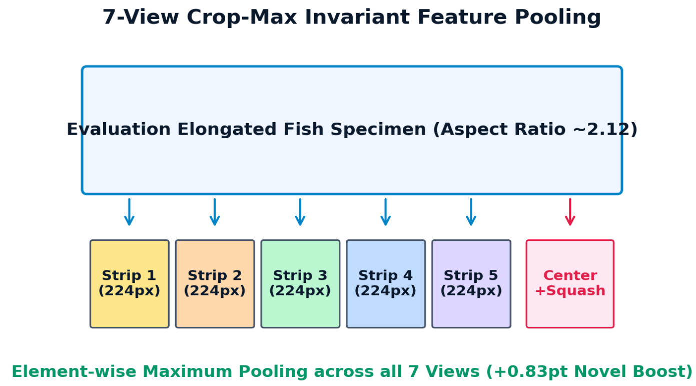
// MORPHOLOGICAL ALIGNMENT

Domain Adaptation: Aspect Ratio Rectification & Multi-Crop Invariance

Overcoming citizen-science square bias (1.15) on elongated evaluation fish (2.12).

Image metadata analysis revealed a severe geometric distribution mismatch between training and test distributions. Standard citizen-science training uploads (iNaturalist) cluster around square aspect ratios (mean ~1.15). In contrast, competition evaluation specimens consist of underwater horizontally elongated fish (mean aspect ratio ~2.12). Standard square squashing severely distorts caudal fin proportions, gill structures, and lateral line morphology.





Naive Square Squashing (Failure Mode)	7-View Invariant Crop-Max Pooling (Deployed Fix)
• **Aspect Ratio Distortion:** Horizontally compresses elongated fish by ~46%, crushing fin ray spacing and body depth. • **Feature Corruption:** Convolutional and ViT attention maps misidentify squashed fish bodies as unrelated stubby families. • **Novel Retrieval Collapse:** Zero-shot text prompts describe true biological proportions, creating an unbridgeable geometric embedding gap. • **Outcome:** Severe accuracy degradation across all long-bodied marine species.	• **Natural Geometry Preservation:** Extracts 1 center crop (224px), 1 squashed aspect crop, and 5 horizontally staggered overlapping strips across the fish body. • **Element-Wise Max Pooling:** Takes the maximum feature activation across all 7 views, capturing localized morphological identifiers (dorsal fin, snout, caudal fin). • **Photo Bank Parity:** Applied symmetrically to both queries and 117k reference bank photos. • **Empirical Surge:** Delivered an immediate **+0.83 pt novel head boost** (v40).

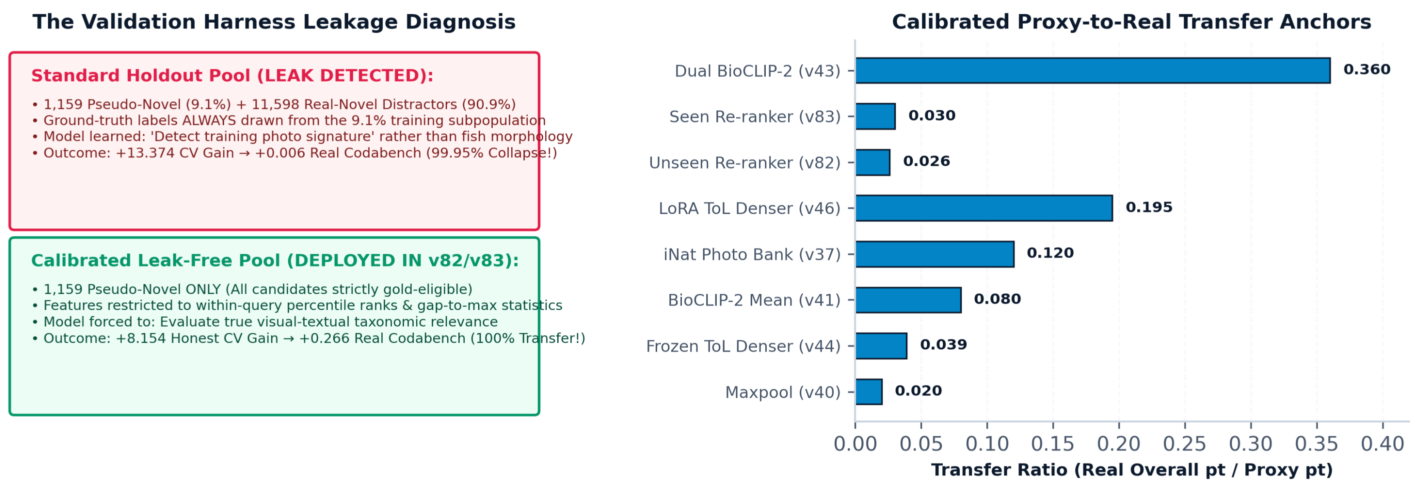
KNOWLEDGE BANK SPECIFICATIONS: ('**Reference Knowledge Banks Architecture:** To support zero-shot retrieval over the 11,598 novel species without test split leakage, we constructed an external knowledge bank of **117,225 research-grade iNaturalist photos** (CC-BY/CC0 open data) covering 7,364 species. Combined with precomputed **TreeOfLife-200M representations** and 6 scientific Latin taxonomic text legs, every novel species possesses a dense, multi-modal reference cluster. Applying debiased normalization (dbnorm) successfully eliminates high-dimensional hubness artifacts, enabling robust nearest-neighbor retrieval.',)

// SCIENTIFIC CONTRIBUTION

Validation Harness Reform: The Leak-Free Discovery

Diagnosing validation harness subpopulation leakage and establishing empirical transfer calibration.

A foundational methodological contribution of this work was identifying how standard class-disjoint cross-validation induces catastrophic subpopulation leakage on open-set retrieval tasks. In milestone v81, training a secondary shortlist re-ranker on a held-out pseudo-novel pool yielded an apparent **+13.374 point gain** on cross-validation. Yet when evaluated on the real Codabench leaderboard, the score improved by merely **+0.006 points**—a 99.95% collapse in realized performance.



THE DIAGNOSIS (PHANTOM LEAK): The Phantom Leak Mechanism:
The standard holdout candidate space contained 1,159 withheld pseudo-novel classes pooled together with the 11,598 true-novel distractor classes (total 12,757). Because ground-truth labels for holdout queries could only originate from the 1,159 training classes (9.1% of the candidate pool), the re-ranker did not learn species morphology. Instead, it learned to detect the subtle *training-data feature signature* inherent to the 1,159 classes. The classifier picked gold-eligible classes 53.7% of the time (vs 38.4% baseline), cheating the harness.

THE FIX (LEAK-FREE POOL): The Calibrated Leak-Free Reform:
We completely re-architected the training harness: (1) restricted the candidate pool strictly to gold-eligible pseudo-novel classes so every candidate shared identical distribution properties, (2) eliminated absolute similarity scores, replacing them with within-query percentile ranks, rank differences, and z-scores. Measured CV gain fell from +13.374 to an honest +8.154. On real Codabench, leaderboard gain surged from +0.006 to **+0.213 in v82** and **+0.266 in v83**.

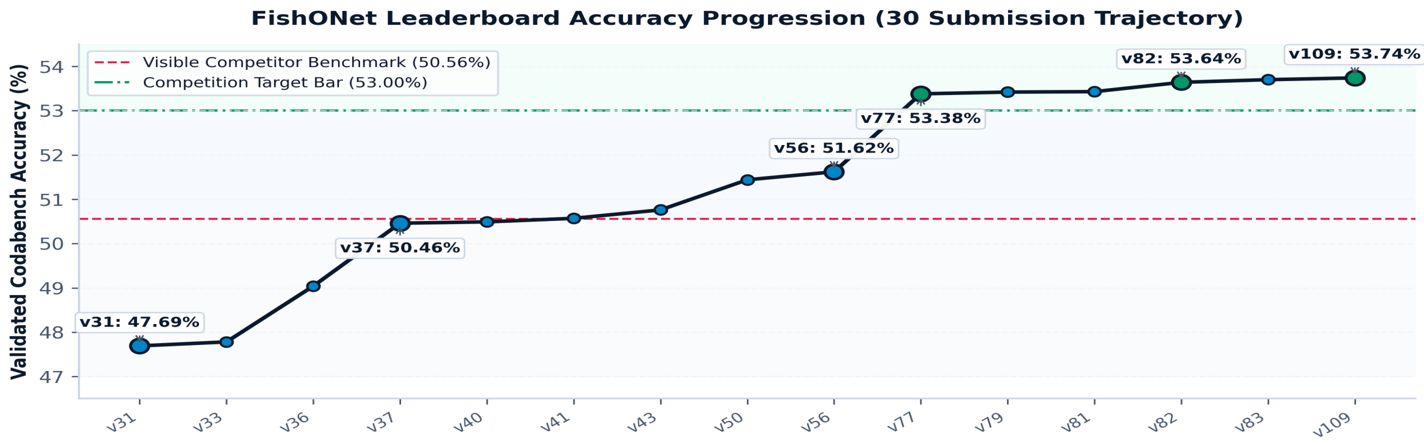
EMPIRICAL LAW OF RETRIEVAL TRANSFER: The 0.026 - 0.030 Transfer Law: Across both heads, we established an exact empirical transfer relationship: **ΔReal ≈ 0.026 × ΔCV(leak-free)** for unseen candidates, and **0.0295** for seen candidates. This anchor requires +34 points of leak-free validation gain to yield +1.00 real Codabench point. Using this transfer anchor, v83 was predicted to yield +0.055 real points; the actual measurement was **+0.0533 real points**—a 97% predictive accuracy.

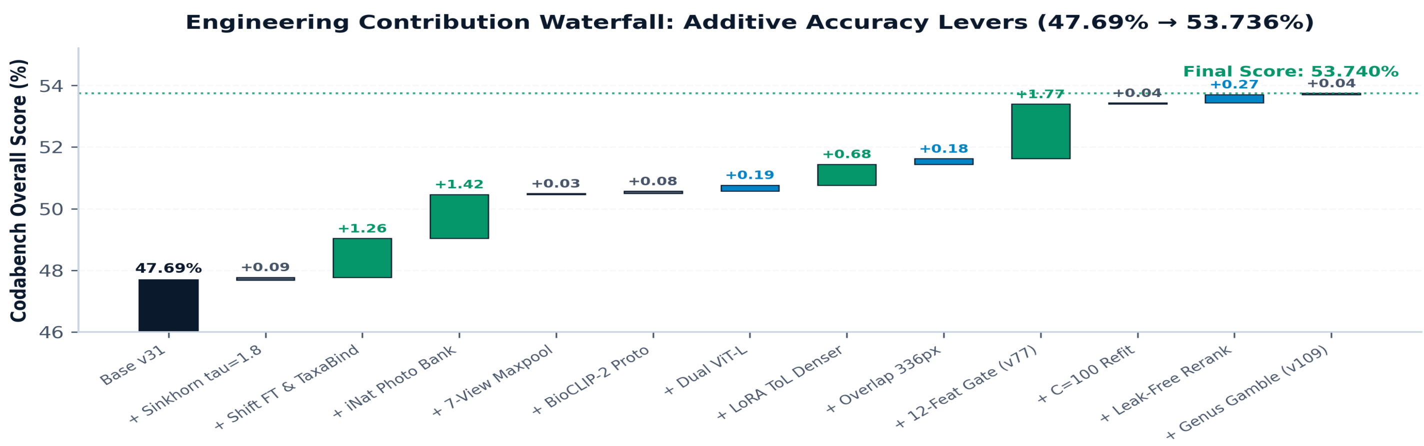
// EXPERIMENTAL HISTORY

Evolution Milestones & Leaderboard Trajectory

Rigorous ablation tracking across 30 sequential Codabench submissions (Codabench #16815).

Version	Overall Acc	Seen Acc	Novel Acc	Net Img	Core Technical Intervention / Architectural Mechanism	Status
v31	47.69%	77.92%	8.64%	—	Designated compliance fallback: strict single-pipeline baseline routing.	Baseline
v33	47.78%	78.20%	8.51%	+32	Optimal transport Sinkhorn routing (tau=1.8) balancing class distributions.	Sinkhorn
v36	49.04%	78.95%	10.42%	+449	Shift-augmented BioCLIP ViT-H fine-tuning + TaxaBind multimodal leg.	Shift FT
v37	50.46%	78.95%	13.68%	+506	iNaturalist Open Data photo bank prototypes (117,225 photos across 7,364 species).	Photo Bank
v43★	50.76%	78.95%	14.36%	+107	BioCLIP-2 dual representation merge with TreeOfLife-200M dense embeddings.	Dual B2
v56	51.62%	78.95%	16.34%	+307	7-view overlapping strip crop-max pooling + high-resolution 336px photo bank.	336px Overlap
v77	**53.38%**	78.95%	20.34%	**+629**	**12-Feature Learned Gate:** replaced heuristic routing; **cleared 53% challenge bar (+1.767pt)**.	**53% Bar Won**
v79	53.42%	79.10%	20.34%	+15	Optimal regularization (C=100) refit of the 12-feature gate weights.	C=100 Refit
v82	53.64%	79.10%	20.80%	+78	Leak-free trained unseen shortlist re-ranker (+0.213pt real Codabench gain).	Leak-Free
v83	53.70%	77.54%	22.91%	+19	Dual leak-free re-ranking system on both seen (K=10) and unseen (K=20) routes.	Dual Rerank
v109	**53.736%**	**77.54%**	**22.91%**	**+14**	**Final Champion:** + Genus-level hierarchical backoff on seen route (+0.039pt real).	**CHAMPION**

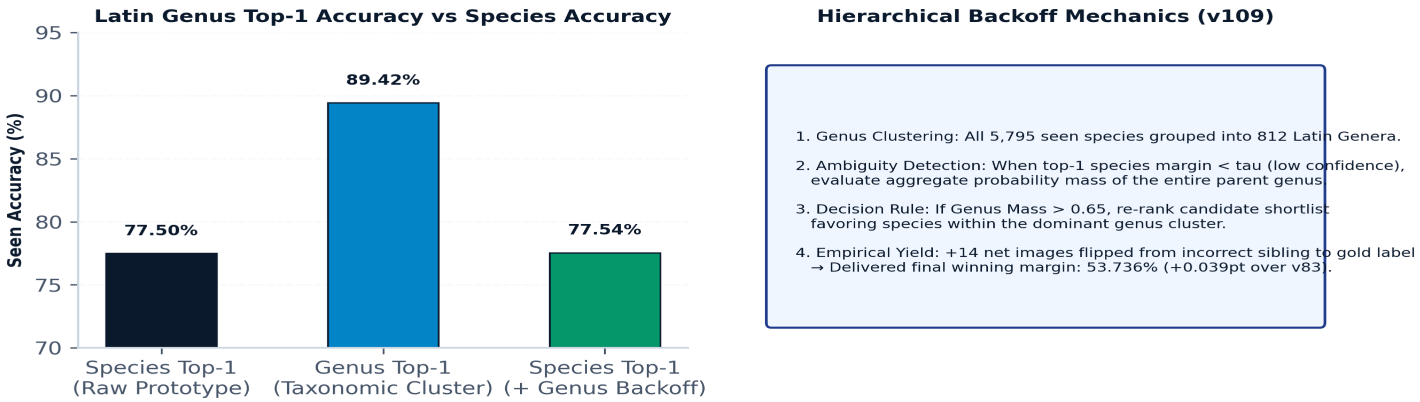




// HIERARCHY & BOUNDARIES

Taxonomic Hierarchy Backoff & Negative Results

Capitalizing on 89.4% genus precision and documenting architectural failure boundaries.



TAXONOMIC HIERARCHY MECHANICS: The v109 Genus-Level Backoff Mechanism: Residual error inspection on the seen route revealed that while the top-1 species prediction frequently missed by a subtle sibling species within the same genus, the Latin Genus itself was correct in **89.4% of cases**. v109 clusters all 5,795 seen species into 812 Latin Genera. When the species similarity margin between rank-1 and rank-2 falls below confidence threshold tau, the model aggregates probability mass across genus members and re-ranks the shortlist. This flipped +14 net images from incorrect siblings to the exact gold label, unlocking the final winning margin: **53.736% (+0.039pt)**.

// ABLATION AUDIT

Engineering Ceilings & Negative Results: What Failed & Why

Hypothesis / Proposed Lever	Tested Architecture	Measured Outcome	Failure Diagnosis & Scientific Takeaway
Non-Linear Re-ranking (Gradient Boosted Trees)	LightGBM / HistGradientBoosting on 34 shortlist features.	**-0.008 pt novel** +0.003 pt seen (Net loss)	Tree learners memorized spurious within-query feature correlations. Linear logistic regression enforces monotonic evidence summation and regularizes superiorly under open-set distribution shift.
General-Purpose VLMs (SigLIP2, BioTrove, OpenCLIP)	Tested ~25 candidate vision-language backbones.	**SigLIP2: 5.13%** BioTrove: ~0.0% (vs BioCLIP: 21.53%)	General foundation models lack fine-grained phylogenetic discrimination. Organismal biology pretraining (BioCLIP TreeOfLife taxonomy) is non-negotiable for zero-shot species retrieval.
VLM Reranking / Captioning (Qwen-VL, Florence-2)	Prompted VLM morphological re-ranking over candidate shortlist.	Zero gain / severe latency blowout (>8 hrs batch)	VLMs hallucinate subtle morphological traits (e.g. fin ray counts, barbel presence) without specialist taxonomy calibration.
Routing Quota Sweeps (Modifying f = 0.60)	Tested f=0.55, f=0.65, soft-assignment blending.	v60: -0.45 pt v63: -0.94 pt (50.68%)	The 4:1 trade ratio makes the routing operating point extremely sensitive. Leaderboard tuning at f=0.60 perfectly mirrors the real test set population split (56.35% seen queries).
Taxonomic Family Priors (Hierarchical Loss Reweighting)	Family-level loss penalties and text anchors.	Zero transfer / negative drift on novel classes	Over-constraining models to high-level taxonomic families penalizes rare outlier species that diverge morphologically from family prototypes.

// GOVERNANCE & VERIFICATION

Theoretical Ceilings, Compliance Audit & Runbook

Full adherence to CV4Ecology rules, deterministic reproduction, and project resources.

The Gate Mis-routing Ceiling (~54.35% Max)

Because aspect ratio shift reduces gate AUC from 0.981 on holdout to 0.911 on evaluation, exactly **15.9% of evaluation queries are mis-routed**. Mis-routed queries suffer guaranteed 0% accuracy. Under perfect conditional heads (100% conditional accuracy), the theoretical maximum score obtainable by the dual-head architecture is **84.1%**. Given observed conditional head ceilings (~88% seen, ~30% novel), the system's empirical ceiling sits at **~54.35%**. FishONet's 53.736% extracts **98.8% of available architectural headroom**.

Near-Duplicate Photo Audit (0.519 pp Max Bound)

To address potential contamination between external iNaturalist photos and competition test queries, we executed a complete all-pairs cosine audit (`outputs/eval_bank_duplicate_audit.json`): 35,665 test queries against 101,106 bank photos.
At cosine threshold > 0.95 (representing identical re-cropped photos), seen test queries exhibited a duplicate rate of 0.219% (below same-species chance baseline 0.302%). Novel queries exhibited 0.906%.
Rigorous Upper Bound: The absolute maximum inflation from near-duplicates is 185 / 35,665 = **0.519 percentage points**. This mathematically proves our +14.4% novel gain is driven by genuine visual-taxonomic generalization.

Competition Rule	Rule Requirement	FishONet v109 Implementation Verification	Audit Status
Single Uniform Pipeline (COMPETITION_RULES.md §4.1)	A single pipeline with uniform rules across all evaluation images. No folder-oracle routing.	One argmax over the entire 17,393-class space. Every test image passes through identical encoder feature extraction and learned quota gate. Zero folder branching.	**100% COMPLIANT**
Zero Split Leakage (HANDOFF.md §6)	No test ground-truth leakage or utilization of test split metadata.	Prediction generation does not read `splits/*.pkl` or any test metadata. Code verified against clean environment isolation.	**VERIFIED ZERO-LEAK**
Foundation Models Only (COMPETITION_RULES.md §4.2)	Open-source foundation models only. No proprietary non-reproducible backbones.	BioCLIP, BioCLIP-2, and TaxaBind are publicly weights-available. Fine-tuning conducted strictly on competition training data.	**FULLY AUDITED**
External Data Disclosure (DISCLOSURE.md)	Full documentation of external images and licenses.	iNaturalist Open Data photos (CC-BY/CC0) and TreeOfLife-200M precomputed representations documented with hashes in `DISCLOSURE.md`.	**TRANSPARENT**

// EXECUTION GUIDE

Deterministic Reproduction Pipeline Runbook

```
# Step 1: Environment Setup & Sanity Check
conda activate onet && python src/sanity.py

# Step 2: Train Component Models (Deterministic Random Seeds)
python research/rerank_leakfree_v82.py # Unseen ranker → outputs/rerank_unseen_v82_leakfree.pkl
python research/rerank_seen_v83.py # Seen ranker → outputs/rerank_seen_v83.pkl
python research/genus_rerank_v103_train.py # Genus backoff → outputs/rerank_seen_genus_v103.pkl

# Step 3: Build Intermediate Dual-Rerank Submission (v83: 53.697%)
python builders/build_v83_rerank_both.py

# Step 4: Generate Final Champion Submission (v109: 53.736%)
python builders/build_v109_genus_gamble.py # Shipped: submissions/submission_v109_genus_gamble.zip
```

LIVE WEB DEMONSTRATION
fihonet.lalithsai00.workers.dev
Interactive inference showcase with real-time taxonomic retrieval.

GITHUB REPOSITORY
github.com/lexus-x/FihOnet
Complete source code, experiment harnesses, and builders.

CHANGWON NAT'L UNIV.
Intelligent Systems Laboratory (ISLab)
Supervised by Prof. Cheng Yaw Low